

Ensuring the Security of Data Generated by IoT Devices



The Internet of Things may have made our lives easier, but the flip side is the risk of losing data and privacy.

However, technology has helped to identify the threats to data with IoT implementation. And there are solutions to these, as discussed in this article.

The Internet of Things or IoT has made life much easier for many of us.

The wireless transfer and connection of data could never have been possible without IoT. In a few years, it is expected that artificial intelligence will make IoT even more independent of human intervention, as devices will be able to communicate with each other.

However, the privacy and confidentiality of the data being processed remains a question mark. Anyone can eavesdrop on this data, intentionally or unintentionally. Devices are interconnected with different hardware and software in IoT, so there are possibilities of confidential

information being leaked by unauthorised sources. All devices send information to private users without encryption, including identification details, location, date of birth, and so on. Privacy is a big concern not only at the business and corporate levels, but also at the state level. Hence, there is a need to focus more on the security aspect when designing and launching IoT based products.

Challenges

Associating an identity with a certain person can lead to profiling and tracking. One of the biggest challenges therefore is to disallow certain IoT operations. As systems attempt to determine and record the position of

individuals across time and space, localisation is another threat. One of the major challenges of IoT security solutions is to develop protocols that prohibit such behaviour for IoT interactions. In e-commerce applications, profiling personal information to infer preferences through similarities with other profiles and data is very common.

For securing sensitive information, the very first step is to provide tangible and physical security to the devices. This includes sensor security, sensor interference, and interception of signals. Without physical security, all other measures will have no positive impact. Sensors and transmission systems need to be safe.

Solutions available

Solutions to these challenges are being proposed by experts, but their effectiveness will depend on proper implementation.

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